

Name: _____

Date: _____

1. Approximate the definite integral using the trapezoidal rule with $n = 6$. Approximate each function value to four decimal places and round the final answer to two decimal places.

$$\int_1^4 \frac{1}{x} dx$$

2. Evaluate.

$$\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} dx$$

3. Evaluate.

$$\int_0^{\pi/2} \cos x \sqrt{3 + 5 \sin x} dx$$

4. Evaluate.

$$\int \frac{d}{dx} \left(\sqrt[5]{x^4 + 2x^2 + 1} \right) dx$$

5. Solve the differential equation subject to the given conditions.

$$\frac{d^2 y}{dx^2} = 6x - 4, \quad y(2) = 4, \quad y'(2) = 5.$$

6. Let $f(x) = 9 - x^2$ for $-2 \leq x \leq 3$, and let P be the regular partition of $[-2, 3]$ into five subintervals. Find the Riemann sum R_P if f is evaluated at the midpoint of each subinterval of P .

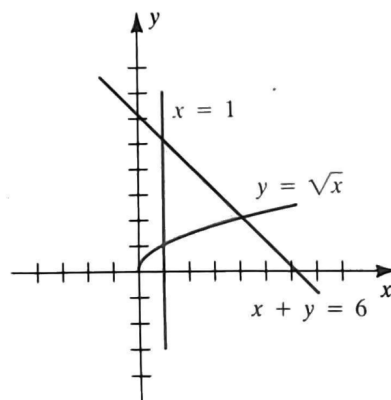
7. Express as one integral.

$$\int_c^e f(x) dx + \int_a^b f(x) dx - \int_c^b f(x) dx - \int_d^d f(x) dx$$

8. Approximate the definite integral using the trapezoidal rule with $n = 5$. Use function values rounded to four decimal places and round the final answer to two decimal places.

$$\int_0^{10} \sqrt{1+x^4} dx$$

9. Set up an integral that can be used to find the area of the region bounded by the three labeled curves.



10. Sketch the region bounded by the graphs of the equations and find its area.

$$y = x^2, \quad y = 4x$$

- 11.** Sketch the region bounded by the graphs of the equations and find its area.

$$y = \frac{1}{x^2}, \quad y = -x^2, \quad x = 1, \quad x = 2$$

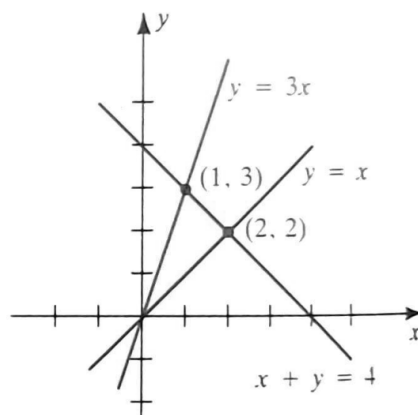
- 12.** Sketch the region bounded by the graphs of the equations and find its area.

$$y = 4 + x, \quad y^2 + x = 2$$

- 13.** Sketch the region bounded by the graphs of the equations and find its area.

$$y = x\sqrt{4 - x^2}, \quad y = 0$$

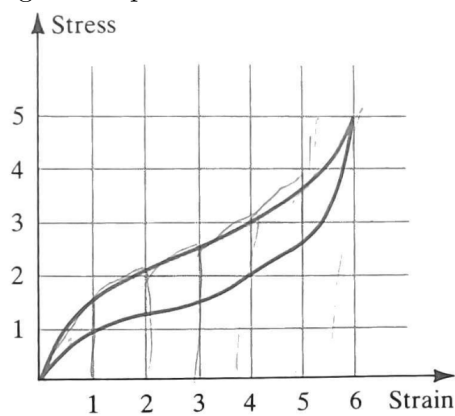
- 14.** Set up sums of integrals that can be used to find the area of the region enclosed by the three lines shown, integrating with respect to **(a)** x and **(b)** y .



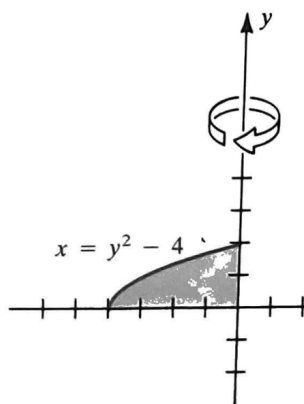
- 15.** Set up integrals, or sums of integrals, that can be used to find the area of the region bounded by the graphs, integrating with respect to **(a)** x and **(b)** y .

$$y = x + 3, \quad x = -y^2 + 3$$

- 16.** The figure shows a stress–strain diagram. Estimate the y -coordinates and approximate the area enclosed by the hysteresis loop using the trapezoidal rule with $n = 6$.



17. Set up an integral that can be used to find the volume of the solid obtained by revolving the shaded region about the indicated axis.



18. Sketch the region R bounded by the graphs of the equations. Find the volume of the solid generated if R is revolved about the x -axis.

$$y = \sqrt{x}, \quad x = 4, \quad y = 0$$

19. Sketch the region R bounded by the graphs of the equations. Find the volume of the solid generated if R is revolved about the x -axis.

$$y = x^2 - 4x, \quad y = 0$$

20. Sketch the region R bounded by the graphs of the equations. Find the volume of the solid generated if R is revolved about the x -axis.

$$y = x^3, \quad x = -2, \quad y = 0$$

21. Sketch the region R bounded by the graphs of the equations. Find the volume of the solid generated if R is revolved about the y -axis.

$$x = 4y - y^2, \quad x = 0$$

Tommy's Questions

Tommy 1. A function F satisfies

$$F'(x) = x\sqrt{1+x^2} \quad (0 \leq x \leq 2), \quad F(0) = 2.$$

- (a) Find an exact formula for $F(x)$.
- (b) Find the average value of F' on $[0, 2]$.
- (c) Use the trapezoidal rule with four equal subintervals to approximate $F(2)$. Round to three decimal places.
- (d) Is the approximation in part (c) greater than or less than the exact value? Justify your answer.

Tommy 2. Let R be the region bounded by

$$y = 1, \quad y = \sqrt{1 + \frac{x^{2026}}{x^{2026} + (1-x)^{2026}}}, \quad x = 0, \quad x = 1.$$

Find the exact volume of the solid obtained by revolving R about the x -axis.

Tommy 3. For each $x \in [0, 2]$, let $f(x)$ be the unique nonnegative number satisfying

$$[f(x)]^5 + f(x) = x.$$

Find the exact volume of the solid obtained by revolving the region bounded by $y = f(x)$ and $y = x/2$ about the y -axis.

Answer Key

1. 1.41.

2. $-\frac{2}{1+\sqrt{x}} + C.$

3. $\frac{32\sqrt{2} - 6\sqrt{3}}{15}.$

4. $\sqrt[5]{x^4 + 2x^2 + 1} + C.$

5. $y = x^3 - 2x^2 + x + 2.$

6. $R_P = \frac{135}{4}.$

7. $\int_a^e f(x) dx.$

8. 341.36.

9. $\int_1^4 (6 - x - \sqrt{x}) dx.$

10. $A = \frac{32}{3}.$

11. $A = \frac{17}{6}.$

12. $A = \frac{125}{6}.$

13. $A = \frac{16}{3}$ (both regions together).

14. (a) $\int_0^1 (3x - x) dx + \int_1^2 (4 - x - x) dx.$

(b) $\int_0^2 \left(y - \frac{y}{3}\right) dy + \int_2^3 \left(4 - y - \frac{y}{3}\right) dy.$

15. (a) $\int_{-6}^{-1} (x + 3 + \sqrt{3 - x}) dx + \int_{-1}^3 2\sqrt{3 - x} dx.$

(b) $\int_{-3}^2 [(3 - y^2) - (y - 3)] dy.$

Answer Key (continued)

16. Approximately 4.2 square coordinate units, using the following estimated readings:

x	0	1	2	3	4	5	6
Upper curve	0	1.5	2.0	2.5	3.0	3.6	5.0
Lower curve	0	1.0	1.3	1.5	2.0	2.6	5.0

Reasonable graph readings give slightly different answers.

17. $V = \pi \int_0^2 (4 - y^2)^2 dy.$

18. $V = 8\pi.$

19. $V = \frac{512\pi}{15}.$

20. $V = \frac{128\pi}{7}.$

21. $V = \frac{512\pi}{15}.$

Tommy 1.

(a) With $u = 1 + x^2$ and $du = 2x dx$,

$$F(x) = \frac{1}{3}(1 + x^2)^{3/2} + C.$$

Since $F(0) = 2$, $C = 5/3$. Thus $F(x) = \frac{(1 + x^2)^{3/2} + 5}{3}.$

(b) The average value is

$$\frac{1}{2} \int_0^2 x \sqrt{1 + x^2} dx = \frac{1}{2} \left[\frac{(1 + x^2)^{3/2}}{3} \right]_0^2 = \frac{5\sqrt{5} - 1}{6}.$$

(c) Write $g(x) = F'(x)$. With $\Delta x = 1/2$,

$$T_4 = \frac{1}{4} [g(0) + 2g(1/2) + 2g(1) + 2g(3/2) + g(2)] \approx 3.4567.$$

Therefore $F(2) \approx F(0) + T_4 = \boxed{5.457}.$

(d) $g''(x) = \frac{x(3 + 2x^2)}{(1 + x^2)^{3/2}} > 0$ for $0 < x \leq 2$. The graph of g is concave up, so the trapezoids overestimate its integral. Adding $F(0)$ preserves the inequality: the approximation is **greater** than the exact value.

Tommy's Questions: Worked Solutions

Tommy 2. The outer and inner radii are

$$R(x) = \sqrt{1 + \frac{x^{2026}}{x^{2026} + (1-x)^{2026}}}, \quad r(x) = 1.$$

The washer formula gives

$$V = \pi \int_0^1 [R(x)^2 - r(x)^2] dx = \pi \int_0^1 \frac{x^{2026}}{x^{2026} + (1-x)^{2026}} dx.$$

Let the last integral be I . Substituting $u = 1 - x$ gives

$$I = \int_0^1 \frac{(1-u)^{2026}}{u^{2026} + (1-u)^{2026}} du.$$

Adding the two expressions yields

$$2I = \int_0^1 \frac{x^{2026} + (1-x)^{2026}}{x^{2026} + (1-x)^{2026}} dx = 1.$$

Thus $\boxed{V = \pi/2}$. The radii must be squared separately; the square of their difference is not a washer area.

Tommy 3. The implicit equation gives $x = y^5 + y$ directly. The line is $x = 2y$. Their intersections satisfy

$$y^5 + y = 2y \implies y(y^4 - 1) = 0.$$

On the nonnegative domain, the intersections are $y = 0$ and $y = 1$, corresponding to $(0, 0)$ and $(2, 1)$. For $0 \leq y \leq 1$, $y^5 \leq y$, so

$$y^5 + y \leq 2y.$$

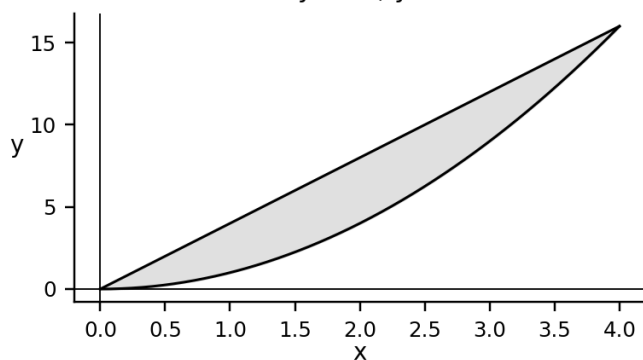
A horizontal slice therefore runs from $x = y^5 + y$ to $x = 2y$. Rotating about the y -axis produces a washer with outer radius $2y$ and inner radius $y^5 + y$:

$$\begin{aligned} V &= \pi \int_0^1 [(2y)^2 - (y^5 + y)^2] dy \\ &= \pi \int_0^1 (3y^2 - 2y^6 - y^{10}) dy \\ &= \pi \left[y^3 - \frac{2}{7}y^7 - \frac{1}{11}y^{11} \right]_0^1 = \boxed{\frac{48\pi}{77}}. \end{aligned}$$

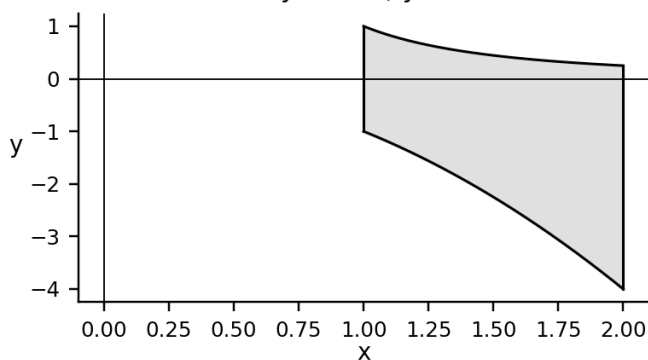
There is no need to solve the fifth-degree equation for $f(x)$.

Answer Key: Region Sketches

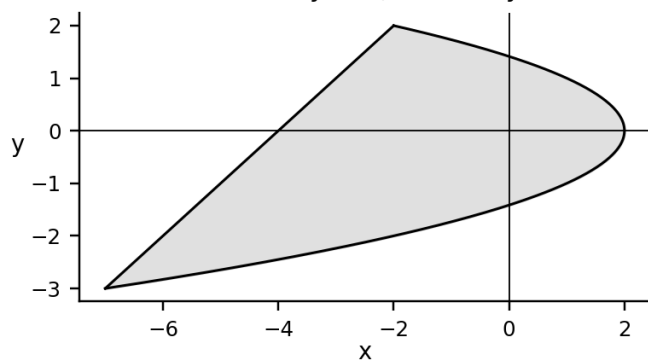
10. $y = x^2$; $y = 4x$



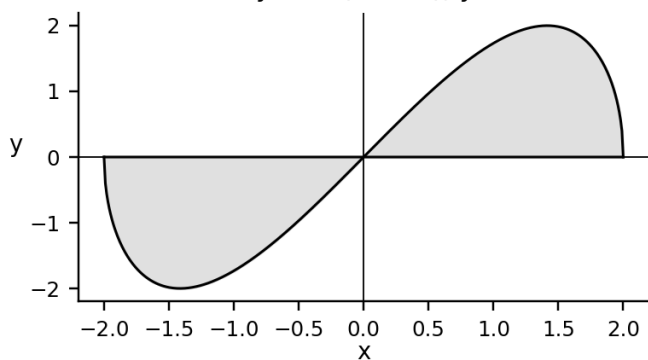
11. $y = 1/x^2$; $y = -x^2$



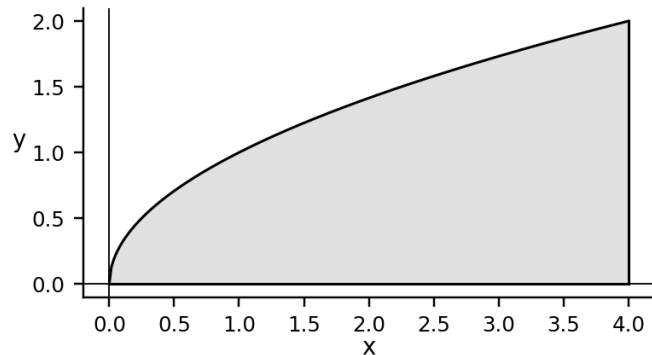
12. $x = y - 4$; $x = 2 - y^2$



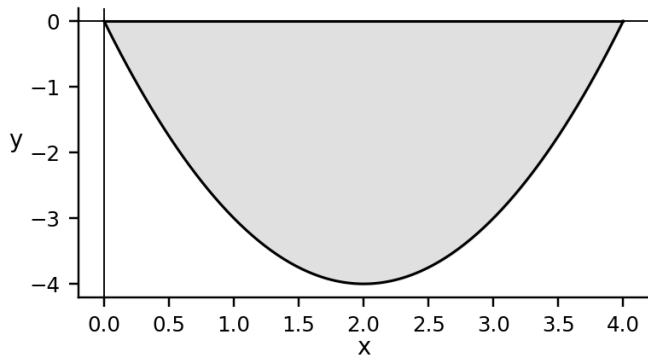
13. $y = x\sqrt{4 - x^2}$; $y = 0$



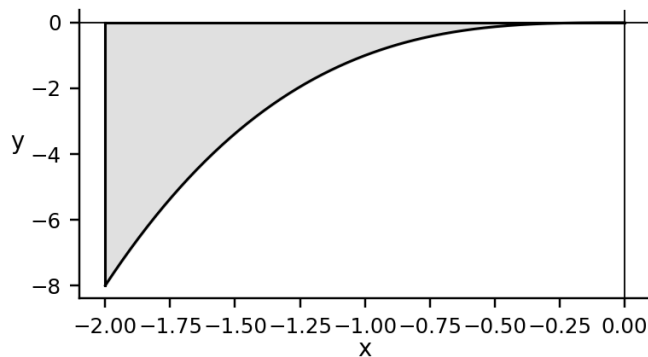
18. $y = \sqrt{x}$; $x = 4$; $y = 0$



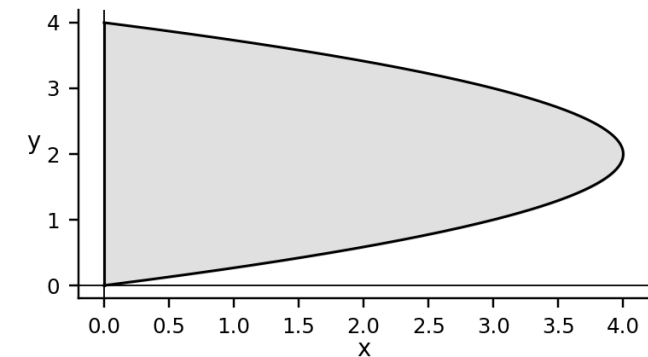
19. $y = x^2 - 4x$; $y = 0$



20. $y = x^3$; $x = -2$; $y = 0$



21. $x = 4y - y^2$; $x = 0$



Where These Questions Came From

These questions come from Sections 5.7, 5.8, 6.1, and 6.2. Try the surrounding problems for more practice.

Guide question	Section	Exercise
1	5.7	5(a)
2	5.8	9
3	5.8	35
4	5.8	39
5	5.8	43
6	5.8	45
7	5.8	49
8	5.8	53(a)
9	6.1	2
10	6.1	5
11	6.1	9
12	6.1	13
13	6.1	21
14	6.1	25
15	6.1	29
16	6.1	39(a)
17	6.2	2
18	6.2	6
19	6.2	7
20	6.2	8
21	6.2	11
Tommy 1	Original — not in the textbook	
Tommy 2	Original — not in the textbook	
Tommy 3	Original — not in the textbook	